

Machine Instruction (EXE) | cop | operands |

- ↳ Arith or logic (add / addi / sub / sh, ...)
- ↳ Load / store (lw / lb / sb / sh / ...)
- ↳ control (Branch) via Label @ memory

Conditional

! conditional

Test

Loop

goto label

Function calls (procedure)

Ex 11

Ex (Loop)

beq $t_0, x_0, Else$ (if $t_0 = x_0$ jump Else)
 $\leftarrow PC = PC + 4$
 if $t_0 \neq x_0$
 Else: ...
 $PC = Else$

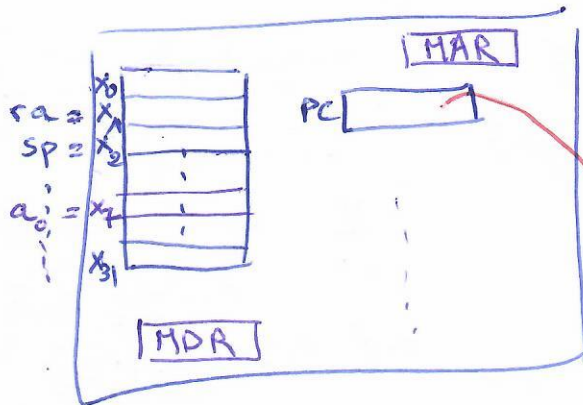
Loop: beq t_0, t_1, End ^{goto}
 $PC + 4$
 $PC = End$

~~PC = End~~ or ~~ret~~ or ~~jalr $x_0, 0(ra)$~~

Jal $x_0, Loop$ $PC = Loop$

End:

CPU

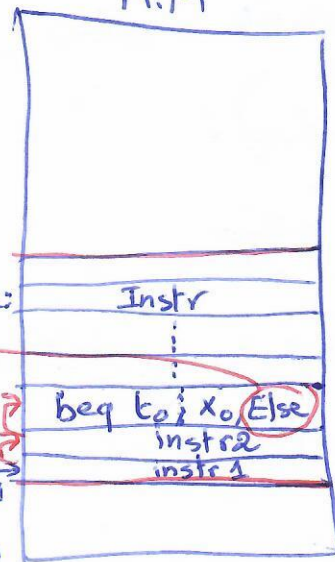


M.M

if $t_0 = 0$

PC = 0 → Else:

PC + 4
 PC = Instr₁



text
 ↑
 Pgm